Thermal effects and energy budgets

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a temperature change only after stating how much energy enters the material.

SUCCESS CRITERIA - I can...

1. Apply an explicit absorbed-energy fraction.
2. Use mass and specific heat in a temperature estimate.
3. State that you are treating phase change and energy loss as zero.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “absorbed” and “specific heat” to explain your answer. Sentence starters are optional.

absorbed: taken into the specified object or system

specific heat: energy per mass needed for a one-degree temperature change

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

A refrigerator uses **15 J** electrical work while moving heat from its colder interior to a warmer room. **Explain why this** is not spontaneous cold-to-hot heat flow.

Word help: spontaneous = able to occur without the specified external work input

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

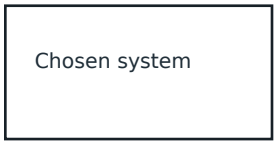
Reference: $Q_{absorbed} = mc\,\Delta T$; $\Delta T = Q_{absorbed}/(mc)$.
First name the absorbed energy. Use only when there is no phase change and c is approximately constant.

Worked reasoning: A surface receives **500 J**. An energy account includes **200 J** reflected, **250 J** absorbed as heating and **50 J** transmitted. The full account sums to **500 J**.

Your turn: Draw proportional energy arrows and **identify which part can directly** raise this surface's temperature.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.



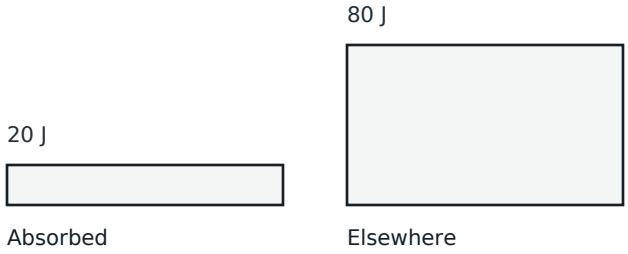
Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A simulated source releases **100 J**. Only **20%** enters a sample. **Draw the energy partition** and **calculate energy absorbed**.

Sentence starter: I found … [unit]. This value means … in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

In a heating simulation, a **1 kg** sample with c = **10 J**/(kg·°C) absorbs **20 J**. Assume no phase change or loss. **Label absorbed energy** on the diagram and calculate ΔT.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A temperature prediction exceeds a material's melting point, but the calculation used only mc ΔT. What must be added or reconsidered?

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A device receives **60 J** and delivers **42 J** useful mechanical energy. **Find other energy** and the useful fraction. **Show the energy changes** and transfers; **label units**.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

If absorbed energy doubles for the same mass and material with no phase change, compare temperature rises. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two surfaces absorb **250 J** but have different masses of the same material.
1. **Compare their temperature rises** if one mass is twice the other.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3